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## Sigma 9 (Web Development)

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### 19. JavaScript (Part 4)

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<> index.html

JS app.js



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JS app.js

1



## for loop

```
for(initialisation; condition; updation) {  
    //do something  
}
```

```
for (let i=1; i<=5; i++) {  
    console.log(i);  
}
```



<> index.html

JS app.js



www.apnacollege.in – To exit full screen, press Esc

JS app.js > ...

```
1 console.log("1");
2 console.log("2");
3 console.log("3");
4 console.log("4");
5 console.log("5");
6
7 for(let i=1; i<=10; i++){
8   console.log(i);
9 }
```



# for loop

```
for(initialisation; ending condition; updation) {  
    //do something  
}
```

```
    ①  
    ↓  
for (let i=1; i<=5; ④ i++) {  
    ③ → console.log(i);  
}
```

output  
1

DRY RUN ✓  
i=1 *tone*  
i=2

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# for loop

```
for(initialisation; ending condition; updation) {  
    //do something  
}
```

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①  
↓  
for (let i=1; i<=5; i++) {  
 ③ → console.log(i);  
 ② ⑤  
 ↓  
 ④  
}

output

1

2

3

4

5

DRY RUN ✓

i=1 true

i=2 true

i=3 true

i=4 true

i=5 true

i=6 False

APNA  
COLLEGE





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JS app.js

×

JS app.js > ...

```

1  console.log("1");
2  console.log("2");
3  console.log("3");
4  console.log("4");
5  console.log("5");
6
7  // for(let i=1; i<=10; i++) {
8  //     console.log(i);
9  // }
10
11  for(let i=10; i>=1; i = i-3) {
12  |   console.log(i);
13  | }
14  let i = 5;
15  console.log(i);

```



# Print all odd numbers (1 to 15)

$i++$   $+1$   
 $i--$   $-1$

start → 1 3 5 7 9 11 13 15 → end

↑ stack update

```
for ( let i=1; i<=15; i=i+2 ) {  
  console.log(i);  
}
```



## Print all even numbers (2 to 10)

2 ← start

4

6

8

10 ← end

```
for ( let i = 2 ; i <= 10 ; i = i + 2 ) {  
    console.log(i);  
}
```





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JS app.js



JS app.js > ...

```

1  // for(let i=1; i<=15; i=i+2) {
2  //     console.log(i);
3  // }
4
5  // console.log("backwards");
6
7  // for(let i=15; i>=1; i=i-2) {
8  //     console.log(i);
9  // }
10
11 for(let i=2; i<=10; i=i+2) {
12 |     console.log(i);
13 | }
14 console.log("backwards");
15
16 for(let i=10; i>=2; i=i-2) {
17 |     console.log(i);
18 | }

```



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02:51 / 03:46



# Print the **multiplication table** for 5

start → 5  
10  
15  
20  
25  
30  
35  
40  
45  
end → 50

+5  
+5  
+5

```
for (let i = 5; i <= 50; i++)
```

||

JS



&lt;&gt; index.html

JS app.js

JS app.js &gt; ...

```
1 let n = prompt("write your number");
2 n = parseInt(i)
3
4 for(let i=n; i<=n*10; i=i+n) {
5     console.log(i);
6 }
7
8 // for(let i=4; i<=40; i=i+4) {
9 //     console.log(i);
10 // }
```

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<> index.html

JS app.js



JS app.js > ...

```

1  let n = prompt("write your number");
2  n = parseInt(n);
3
4  for(let i=n; i<=n*10; i=i+n) {
5      console.log(i);
6  }
7
8  // for(let i=4; i<=40; i=i+4) {
9      // console.log(i);
10 // }

```

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## Nested for loop

```
for (let i=1; i<=3; i++) {  
    for (let j=1; j<=3; j++) {  
        console.log(j);  
    }  
}
```





# Nested for loop

```
for (let i=1; i<=3; i++) {  
  for (let j=1; j<=3; j++) {  
    console.log(j);  
  }  
}
```

outer loop i=1  
j=1  
j=2  
j=3  
j=4

outer loop i=2  
j=1  
j=2  
j=3  
j=4

outer loop i=3  
j=1  
j=2  
j=3  
j=4

outer loop i=4

output  
APNA COLLEGE  
1  
2  
3  
1  
2  
3  
1  
2  
3



## while loop

```
while (condition) {  
    //do something  
}
```

```
let i = 1;  
  
while (i <= 5) {  
    console.log(i);  
    i++;  
}
```



# while loop

```
while true (condition) {  
  //do something  
}
```

```
let i = 1;
```

```
while (i <= 5) {
```

```
  console.log(i); ✓
```

```
  i++; ✓
```

```
  }  
✓
```

i = 1 (T)

i = 2 (T)

i = 3 (T)

i = 4 (T)

i = 5 (T)

i = 6 (F)

1  
2  
3  
4  
5

APNA  
COLLEGE



&lt;&gt; index.html

JS app.js



JS app.js &gt; ...

```
1  // let i=5;
2
3  // while(i>=1) {
4  //     console.log(i);
5  //     i--;
6  // }
7
8  let i=0;
9
10 while(i<=20) {
11     console.log(i);
12     i++;
13 }
```



## Favorite Movie ✓

```
let favMovie = "Avatar";  
let guess = prompt(" ");
```





## Favorite Movie ✓

```
let favMovie = "Avatar";
```

```
let guess = prompt("");
```

```
while((guess != favMovie) && (guess != 'quit')) {
```

```
    console.log("wrong");
```

```
    guess = prompt("");
```

```
}
```

2 condition

1) right guess  $\Rightarrow$  guess = favMovie

2) guess = "quit"



<> index.html

JS app.js

JS app.js > ...

```

1  const favMovie = "avatar";
2
3  let guess = prompt("guess my favorite movie");
4
5  while( (guess !== favMovie) && (guess !== "quit")) {
6      |   guess = prompt("wrong guess. please try again");
7  }
8
9  if(guess == favMovie) {
10     |   console.log("congrats!!");
11 } else {
12     |   console.log("quit")
13 }

```

log(...data: any[]): void



&lt;&gt; index.html

JS app.js



JS app.js &gt; ...

```
8
9  // if(guess == favMovie) {
10 //     console.log("congrats!!");
11 // } else {
12 //     console.log("you quit")
13 // }
14
15
16 let i=1;
17 while(i<=5) {
18     if(i == 3) {
19         break;
20     }
21     console.log(i);
22     i++;
23 }
```



## break keyword

loop execution stop



# Loops with Arrays

```
let fruits = ["mango", "apple", "banana", "litchi", "orange"];

for(let i=0; i<fruits.length; i++) {
  console.log(i, fruits[i]);
}
```





# Loops with Arrays

```
let fruits = ["mango", "apple", "banana", "litchi", "orange"];  
for(let i=0; i<fruits.length; i++) {  
  console.log(i, fruits[i]);  
}
```

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|     |   |        |
|-----|---|--------|
| i=0 | 0 | mango  |
| i=1 | 1 | apple  |
| i=2 | 2 | banana |
| i=3 | 3 | litchi |
| i=4 | 4 | orange |



&lt;&gt; index.html

JS app.js



JS app.js &gt; ...

```
1 let fruits = ["mango", "apple", "banana", "litchi", "orange"];
2
3 for(let i=0; i<fruits.leng let i: number
4   console.log(i, fruits[i]);
5 }
```

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04:06 / 07:00



&lt;&gt; index.html

JS app.js



JS app.js &gt; ...

```
1  let fruits = ["mango", "apple", "banana", "litchi", "orange"];
2  fruits.push("pineapple");
3
4  // for(let i=0; i<fruits.length; i++) {
5  //     console.log(i, fruits[i]);
6  // }
7
8  for(let i=fruits.length-1; i>=0; i--) {
9      console.log(i, fruits[i]);
10 }
```

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06:56 / 07:00



# Loops with Arrays

## Nested Loops with Nested Arrays

```
let heroes = [ ["ironman", "spiderman", "thor"], ["superman", "wonder woman", "flash"]];  
for(let i=0; i<heroes.length; i++) {  
  console.log(`List #${i}`);  
  for(let j=0; j<heroes[i].length; j++) {  
    console.log(heroes[i][j]);  
  }  
}
```

Handwritten annotations: A bracket above the first array ["ironman", "spiderman", "thor"] is labeled "Marvel". A bracket above the second array ["superman", "wonder woman", "flash"] is labeled "DC". A bracket below the inner loop's condition is labeled "1".





&lt;&gt; index.html

JS app.js

JS app.js &gt; ...

```
1  let heroes = [  
2      ["ironman", "spiderman", "thor"],  
3      ["superman", "wonder woman", "flash"]  
4  ]  
5  
6  for(let i=0; i<heroes.length; i++) {  
7      console.log(i, heroes[i], heroes[i].length);  
8      for(let j=0; j<heroes[i].length; j++) {  
9          console.log(`j=${j}, ${heroes[i][j]} `)  
10     }  
11 }
```



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05:09 / 08:32





Loops

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Elements Console Sources

top Filter Default levels

No Issues

```
0 ▶ (3) ['ironman', 'spiderman', 'thor'] 3 app.js:7
j=0, ironman app.js:9
j=1, spiderman app.js:9
j=2, thor app.js:9
1 ▶ (3) ['superman', 'wonder woman', 'flash'] 3 app.js:7
j=0, superman app.js:9
j=1, wonder woman app.js:9
j=2, flash app.js:9
>
```



&lt;&gt; index.html

JS app.js

JS app.js &gt; ...

```
1  // let heroes = [  
2  //     ["ironman", "spiderman", "thor"],  
3  //     ["superman", "wonder woman", "flash"]  
4  // ]  
5  
6  // for(let i=0; i<heroes.length; i++) {  
7  //     console.log(i, heroes[i], heroes[i].length);  
8  //     for(let j=0; j<heroes[i].length; j++) {  
9  //         console.log(`j=${j}, ${heroes[i][j]}`);  
10 //     }  
11 // }  
12  
13 let student = [ ["aman", 95], ["shradha", 94.4], ["karan", 100]];  
14  
15 for(let i=0; i<student.length; i++) {  
16     for(let j=0; j<student[i].length; j++) {  
17         console.log(student[i][j])  
18     }  
19 }
```



Loops

File | /Users/shradhakhapra/WebClassroom/JavaScript/index.html

Elements Console Sources

top Filter Default levels

No Issues

|         |           |
|---------|-----------|
| aman    | app.js:17 |
| 95      | app.js:17 |
| shradha | app.js:17 |
| 94.4    | app.js:17 |
| karan   | app.js:17 |
| 100     | app.js:17 |

>



## for of loop

```
for (element of collection) {  
    //do something  
}
```

```
let fruits = ["mango", "apple", "banana", "litchi", "orange"];  
  
for(fruit of fruits) {  
    console.log(fruit);  
}
```

```
for(char of "apnacollege") {  
    console.log(char);  
}
```





## for of loop

`for (element of collection) {`  
    `//do something`  
}

*Handwritten notes:*  
- A red bracket under "collection" points to "array" and "string".  
- The word "for" is written above "while" with a greater-than sign (>) between them.

```
let fruits = ["mango", "apple", "banana", "litchi", "orange"];
```

```
for(fruit of fruits) {  
    console.log(fruit);  
}
```

```
for(char of "apnacollege") {  
    console.log(char);  
}
```





&lt;&gt; index.html

JS app.js

JS app.js &gt; ...

```
1 let fruits = ["mango", "apple", "banana", "orange", "litchi"];
2
3 for(fruit of fruits) {
4   console.log(fruit);
5 }
6
7 |
```



<> index.html

JS app.js X

JS app.js > let fruits: string[]

```
1 let fruits = ["mango", "apple", "banana", "orange", "litchi"];
2
3 for(fruit of fruits) {
4     console.log(fruit);
5 }
6
7 for(char of "apnacollege") {
8     console.log(char);
9 }
```



## Nested for of loop

```
let heroes = [ ["ironman", "spiderman", "thor"], ["superman", "wonder woman", "flash"]];  
  
for(list of heroes) {  
  for(hero of list) {  
    console.log(hero);  
  }  
}
```



&lt;&gt; index.html

JS app.js



JS app.js &gt; ...

```
1 let hero = ["batman", "wonder woman"], ["spiderman", "ironman", "thor"]];
2
3 for(list of hero) {
4   for(name of list) {
5     console.log(name);
6   }
7 }
```

const name: void

@deprecated

for(name of list) {

console.log(name);



# Todo App

- **list** - to show all todos
- **add** - to add a todo
- **delete** - to delete a task
- **quit** - to exit the todo



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&lt;&gt; index.html

JS app.js



JS app.js &gt; ...

```
5   while(true) {  
6       if(req == "quit") {  
7           console.log("quitting app");  
8           break;  
9       }  
10  
11      if(req == "list") {  
12          console.log("-----");  
13          for(task of todo) {  
14              console.log(task);  
15          }  
16          console.log("-----");  
17      } else if(req == "add") {  
18          let task = prompt("please enter the task you want to add");  
19          todo.push(task);  
20          console.log("task added");  
21      }  
22  }  
23 }
```



&lt;&gt; index.html

JS app.js

JS app.js &gt; ...

```
6     if(req == "quit") {
7         console.log("quitting app");
8         break;
9     }
10
11     if(req == "list") {
12         console.log("-----");
13         for(let i=0; i<todo.length; i++) {
14             console.log(i, todo[i]);
15         }
16         console.log("-----");
17     } else if(req == "add") {
18         let task = prompt("please enter the task you want to add");
19         todo.push(task);
20         console.log("task added");
21     } else if(req == "delete") {
22         let idx = prompt("please enter the task index");
23         todo.splice(idx, 1);
24         console.log("task deleted");
25     }
26
27     req = prompt("please enter your request");
```



Loops

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APNA COLLEGE

elements Console Sources

Filter Default levels

app.js:12  
app.js:16  
app.js:20  
app.js:12  
app.js:14  
app.js:14  
app.js:14  
app.js:16  
app.js:20  
is:12

3 task added  
-----  
0 'eat'  
1 'sleep'  
2 'code'  
-----  
task added  
-----  
0 'eat'  
1 'sleep'  
3 'go for a walk'

# To do App

Enter "list" to show all tasks

Enter "add" to add a task

Enter "delete" to remove a task

Enter "quit" to quit todo

This page says  
please enter your request

Cancel OK



JS app.js &gt; ...

```
1  let todo = [];  
2  
3  let req = prompt("please enter your request");  
4  
5  while(true) {  
6      if(req == "quit") {  
7          console.log("quitting app");  
8          break;  
9      }  
10  
11     if(req == "list") {  
12         console.log("-----");  
13         for(let i=0; i<todo.length; i++) {  
14             console.log(i, todo[i]);  
15         }  
16         console.log("-----");  
17     } else if(req == "add") {  
18         let task = prompt("please enter the task you want to add");  
19         todo.push(task);  
20         console.log("task added");  
21     } else if(req == "delete") {  
22         let idx = prompt("please enter the task index");  
23         todo.splice(idx, 1);  
24         console.log("task deleted");  
25     } else {
```

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## JS (Part 4)

### Practice Questions

Qs1. Write a JS program to delete all occurrences of element 'num' in a given array.

Example : if *arr* = [1, 2, 3, 4, 5, 6, 2, 3] & *num* = 2

Result should be *arr* = [1, 3, 4, 5, 6, 3]

Qs2. Write a JS program to find the no of digits in a number.

Example : if *number* = 287152, *count* = 6

Qs3. Write a JS program to find the sum of digits in a number.

Example : if *number* = 287152, *sum* = 25

Qs4. Print the factorial of a number n.

[Factorial of a number n is the product of all positive integers less than or equal to a given positive integer and denoted by that integer. ]

Example :

7! (factorial of 7) =  $1 \times 2 \times 3 \times 4 \times 5 \times 6 \times 7 = 5040$

5! (factorial of 5) =  $1 \times 2 \times 3 \times 4 \times 5 = 120$

3! (factorial of 3) =  $1 \times 2 \times 3 = 6$

0! Is always 1

Qs5. Find the largest number in an array with only positive numbers.